

Decoys Cannot Go Everywhere

Mapping the deception surface in MITRE ATT&CK

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Research has been focusing on better decoys, interaction, engagement... often assuming a decoy can be placed wherever there is attacker behavior.

But where can decoys really exist?

RQ1: Which attacker behaviors admit a decoy, and for those that do, what signal would those decoys produce?

01 · INFRASTRUCTURE DECEPTION

The attacker must reach something we can fake, instrument, and watch.

02 · MITRE ATT&CK

Evaluated against MITRE ATT&CK (v18.1): 14 tactics, 250 techniques

A four-criterion codebook and scoring rubric

Feasibility

Can we fake it?

Interaction

If we fake it, will
they touch it?

Intelligence Yield

If they touch it, do
we learn anything?

Malice fidelity

If we learn, can we
know it was an
attacker?

Yes · Mostly Yes · Mostly No · No

Brute Force T1110

(...) Without knowledge of the password for an account or set of accounts, an adversary may systematically guess the password using a repetitive or iterative mechanism (...)

Feasibility



Interaction



Intelligence yield



Malice fidelity



Brute Force T1110

(...) Without knowledge of the password for an account or set of accounts, an adversary may systematically guess the password using a repetitive or iterative mechanism (...)

Feasibility

Yes

Interaction

Yes

Intelligence yield

Yes

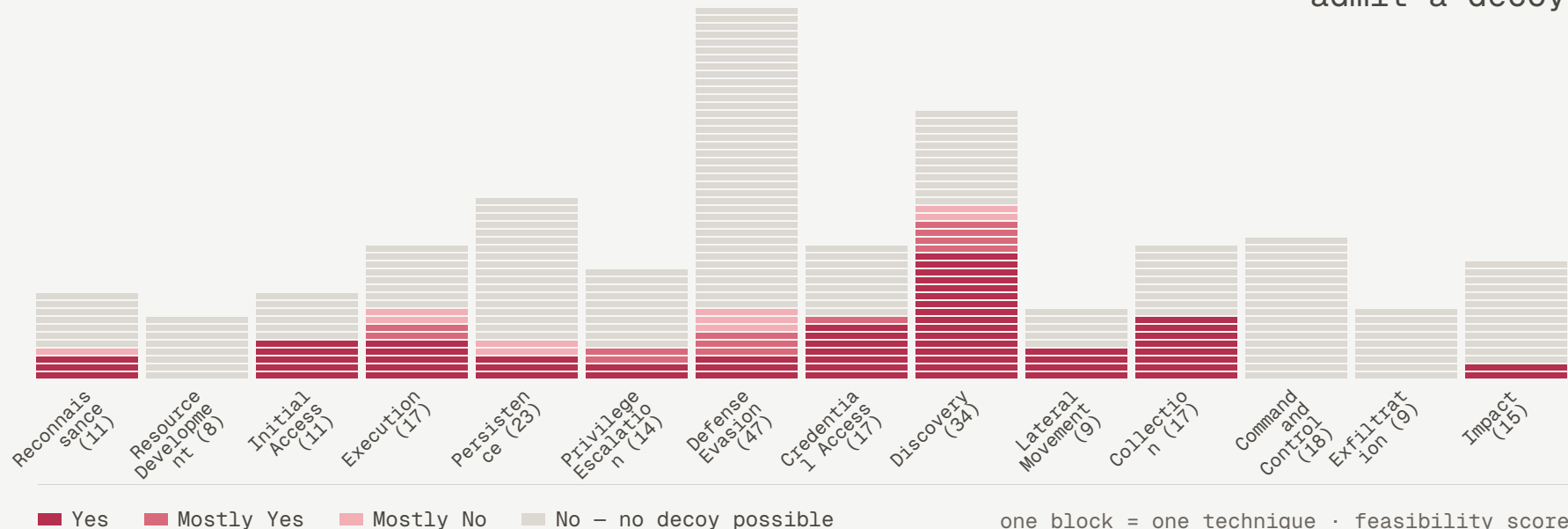
Malice fidelity

Mostly Yes

The deception surface is sparse

80 / 250

32% techniques
admit a decoy



one block = one technique · feasibility score

RQ2: Why do some attacker behaviors admit a decoy and others do not?

Outside defender space

Some techniques happen outside defender-controlled space or depend on attacker-controlled infrastructure,

Resource Development (8) Command and Control (18)
Exfiltration (9)

Inside, but no contact point

In some techniques the attacker never reaches for a defender-controllable resource.

Defense Evasion (38) Persistence (18)
Privilege Escalation (10)

Some behaviors might welcome persona-based or narrative deception, not traditional decoys.

Placement rule

Sweep

The attacker hits everything in range and encounters the decoy as a side effect.

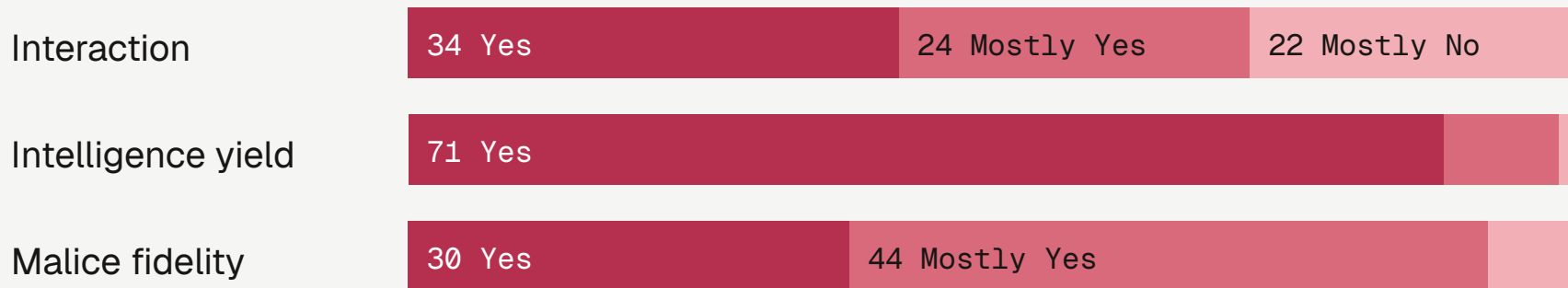
scanning · spraying · collection · ransomware

Seek

the attacker is looking for a specific kind of asset, either to read it or to act on it.

credentials · services · shares · mailboxes

If we have interaction, intelligence is there

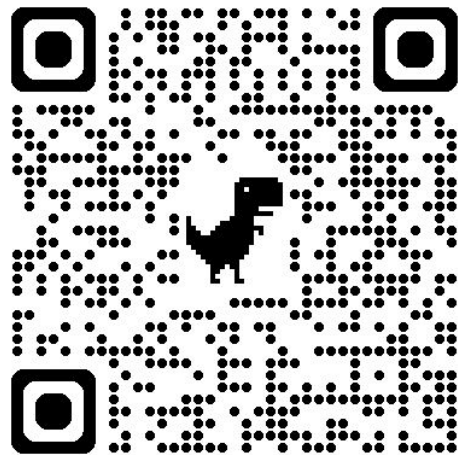


The codebook is usable, but needs diverse insights

32% is just the baseline

We need more people discussing and interpreting feasibility across ATT&CK

Interactive deception explorer →



Perfect decoys in the wrong place **lose all their value.**



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